Everon Unified Field Theory integration of Riemann Hypothesis using BK Hamiltonian

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Abstract

This paper presents an interdisciplinary approach to the Riemann Hypothesis by integrating Everon Quantum Field Theory (EQFT) with spectral analysis techniques. Building upon the Berry–Keating conjecture, we construct a self-adjoint operator whose spectrum aligns with the nontrivial zeros of the Riemann zeta function. By modeling this operator within a curved spacetime framework, specifically employing a massless Dirac fermion in Rindler spacetime with delta function potentials localized at square-free integers, we establish a connection between the distribution of zeta zeros and the geometry of spacetime. Spectral analysis utilizing zeta function regularization and heat kernel methods reveals that the eigenvalues of the constructed operator correspond to the imaginary parts of the nontrivial zeros. This correspondence teaches a link between number theory and quantum physics, implying that the distribution of prime numbers and the zeros of the zeta function are intrinsically connected to the fabric of the universe as described by EQFT. The integration of the Berry–Keating Hamiltonian into the EQFT framework provides a novel perspective on the Riemann Hypothesis, offering pathways for its solution through the lens of quantum field theory and spacetime geometry.

Keywords: Riemann Hypothesis, delta function, Schrodinger equation,

Introduction

Everonic Unified fields

Everon Lagrangian (Full Field Inclusion)

$$\mathscr{L}_{\text{EQFT}} = \mathscr{L}_{\text{grav}} + \mathscr{L}_{\text{spinor}} + \mathscr{L}_{\text{holo}} + \mathscr{L}_{\text{photon}} + \mathscr{L}_{\text{phonon}} + \mathscr{L}_{\text{int}}$$

Encapsulates all quantum, holonomic, acoustic, gravitational, and interaction terms within the Everon field manifold.

Everon Hamiltonian (Full Field Inclusion)

$$\mathscr{H}_{\text{EQFT}} = \mathscr{H}_{\text{grav}} + \mathscr{H}_{\text{spinor}} + \mathscr{H}_{\text{holo}} + \mathscr{H}_{\text{photon}} + \mathscr{H}_{\text{phonon}} + \mathscr{H}_{\text{int}}$$

The total Hamiltonian corresponding to the Lagrangian formulation, expressed via Legendre transformation and supporting holonomic dynamics. Hamiltonian, Lagrangian and Newtonian solutions emerge from the unification of the following theory and quantum field frameworks; we begin with:

1. Seiberg–Witten Equations (Topological Gauge Theory): $D_A\psi = 0 \qquad F_A^+ = \sigma(\psi,\psi)$.

2. Holonomic Gauge Fields: $\mathscr{A}_\mu = \langle\psi\,|\,\mathscr{U}_\gamma^\dagger \nabla_\mu \mathscr{U}_\gamma\,|\,\psi\rangle, \quad F_{\mu\nu}^{(\mathscr{A})} = \partial_\mu \mathscr{A}_\nu - \partial_\nu \mathscr{A}_\mu + [\mathscr{A}_\mu, \mathscr{A}_\nu]$

3. Bogoliubov Transformation (Photonic/Quantum Optics): $a_k' = u_k a_k + v_k b_k^\dagger$

4. Phonon Field Theory (Acoustic Quantum Fields):

$$\mathscr{L}_{\text{phonon}} = \frac{1}{2}\left(\partial^\mu \phi_{\text{ac}} \partial_\mu \phi_{\text{ac}} - V(\phi_{\text{ac}})\right), \quad V(\phi_{\text{ac}}) = \frac{1}{2} m^2 \phi_{\text{ac}}^2 + \lambda \phi_{\text{ac}}^4$$

5. Gravitational Fields (Einstein–Cartan Formalism): $\mathscr{L}_{\text{grav}} = \frac{1}{2\kappa}\sqrt{-g}(R + \Lambda)$

6. Quantum Holonomic Phase Loop Equation: $\phi(t) = \oint_C \mathscr{A}_\mu dx^\mu = \gamma_{\text{hol}}(t)$

7. Newtonian Limit with Holonomic Correction (Acoustic and Navigation-Enabled):

$$m\ddot{x} = -\nabla V_{\text{eff}} + \hbar \frac{d}{dt}(\nabla \gamma_{\text{hol}})$$

8. Transparency Equation (Quantum Field Opacity): $O(x,t) = 1 - \alpha \left| E(x,t) e^{i\gamma_{\text{hol}}(x,t)} \right|^2$

9. Biofield Extraction and Filtering: $|\Phi(t)\rangle = \sum_i c_i(t)\,|f_i\rangle \quad \rightarrow \quad |\Phi'(t)\rangle = \sum_{i\in H} c_i(t)\,|f_i\rangle$

10. Quantum Power Extraction: $P(t) = \sum_n \eta_n \left| E_n(t) e^{i\gamma_n(t)} \right|^2$

11. Elemental Extraction Phase Encoding: $E_{\text{extract}}(x,t) = E_0 \cdot e^{i\gamma_k(x,t)}$

$$\text{where, } \gamma_k(x,t) = \arg(\chi_k(x,t))$$

12. Alcubierre Warp Metric: $ds^2 = -c^2 dt^2 + [dx - v_s(t) f(r_s) dt]^2 + dy^2 + dz^2$

13. Geodesic Equation (Spacetime Trajectories): $\dfrac{d^2 x^\mu}{d\tau^2} + \Gamma^\mu_{\nu\rho} \dfrac{dx^\nu}{d\tau} \dfrac{dx^\rho}{d\tau} = 0$

14. Bogoliubov Metasurface Transformations:

$$\hat{A}_j = \sum_k \left(U_{jk} \hat{a}_k + V_{jk} \hat{a}_k^\dagger \right)$$

$$H = \sum_k \epsilon_k \hat{a}_k^\dagger \hat{a}_k + \sum_k \Delta_k \hat{a}_k^\dagger \hat{a}_{-k}^\dagger + \text{h.c.}$$

$$\Psi_{\text{out}}(x,t) = \mathscr{M}(x,t) \cdot \left(u_k \Psi_{\text{in}} + v_k \Psi_{\text{in}}^* \right)$$

15. S action in Berry--Keating Hamiltonian:

$$S = \int dt\, dx\, \phi^*(x,t) \left(i\hbar \frac{\partial}{\partial t} - \hat{H}_{\text{BK}} \right) \phi(x,t)$$

By unifying these sets of equations, we can freely navigate the frameworks.

We apply Everon Quantum Field Theory (EQFT) to the Riemann Hypothesis, exploring the formulation of equations across four areas: 1) constructing a self-adjoint operator [6], 2) modeling on curved spacetime, 3) spectral analysis, and 4) physical interpretation within Everon Quantum Field Theory.

1. Constructing a Self-Adjoint Operator

Inspired by Berry–Keating conjecture, consider the operator:

$$\hat{H}_{BK} = \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$$

where

$\hat{H}_{BK}$ is Berry–Keating Hamiltonian operator, a symmetrization of classical Hamiltonian $H=xp$, proposed by Berry and Keating, whose spectrum corresponds to nontrivial zeros of the Riemann zeta function.

$\hat{x}$ is the position operator in QM acting by multiplication: $x^{\psi}(x) = x\psi(x)$

$\hat{p}$: The momentum operator in quantum mechanics. In the position representation, it acts as a derivative: $\hat{p}\psi(x) = -i\hbar\frac{d}{dx}\psi(x)$

$\hbar$ is the reduced Planck constant. defined as $\hbar = \frac{h}{2\pi}$ where h is Planck's constant.

$\psi(x)$ is the wavefunction of a quantum system in position space providing probability amplitude for finding a particle at position x. The square modulus $|\psi(x)|^2$ gives the probability density.

i is he imaginary unit, defined by $i^2=-1$, which appears in quantum mechanics due to the complex nature of wavefunctions and operators.

$\frac{d}{dx}$ is the derivative with respect to position x. It represents the rate of change of the wavefunction with position, which is related to the momentum of the particle.

Symmetrization of $\frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$ ensures that the Hamiltonian operator is Hermitian (self-adjoint), which is essential for the physical requirement that observable quantities have real eigenvalues.

The Berry–Keating Hamiltonian is of interest in the context of the Hilbert–Pólya conjecture, which posits that the nontrivial zeros of the Riemann zeta function correspond to the eigenvalues of a self-adjoint operator, where $\hat{x}$ and $\hat{p}$ are the position and momentum operators, respectively. The Berry-Keating Hamiltonian is:

$$\hat{H_{BK}} = \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x}) = -i\hbar\left(x\frac{d}{dx} + \frac{1}{2}\right)$$

This operator is symmetric and, under appropriate domain considerations, can be made self-adjoint. Its spectrum is continuous over the real line, and with suitable boundary conditions or modifications, it can be tailored to yield a discrete spectrum corresponding to the imaginary parts of the nontrivial zeros of the zeta function[1]. Therefore, the operator provided is the explicit form of the Berry–Keating Hamiltonian in the position representation.This operator is significant in theoretical physics because it has been proposed as a candidate for a Hamiltonian whose eigenvalues correspond to the nontrivial zeros of the Riemann zeta function, thus providing a link between quantum mechanics and number theory.

To incorporate this into the Everon Quantum Field Theory (EQFT) framework, we consider the following approach:

We define a scalar field $\phi(x)$ within the EQFT framework.

We construct an action S that includes the Berry–Keating Hamiltonian:

$$S = \int dt\, dx\, \phi^*(x,t)\left(i\hbar\frac{\partial}{\partial t} - \hat{H}_{\mathrm{BK}}\right)\phi(x,t)$$

Quantize the field $\phi(x,t)$ using standard techniques, ensuring that the resulting theory respects the symmetries and structures inherent in EQFT.

Analyze the spectrum of the quantized theory to identify correspondences with the nontrivial zeros of the Riemann zeta function.

where

S is the action functional, a central quantity in both classical and quantum mechanics. It encapsulates the dynamics of a system and is used in the principle of least action, which states that the path taken by the system between two states is the one that minimizes (or makes stationary) the action.

$\int dt\ dx$: denotes integration over time t and space x, summing contributions from all spacetime points.

$\phi(x,t)$ is a complex scalar field representing the quantum state of the system at position x and time t. In quantum field theory, fields like ϕ are fundamental entities, and particles are viewed as excitations of these fields.

$\phi^*(x,t)$ is the complex conjugate of the field ϕ(x,t). When combined as $\phi^*(x,t)$ $\phi(x,t)$, it gives the probability density of finding a particle at position x and time t

i is the imaginary unit, satisfying $i^2=-1$. Its presence in quantum mechanics reflects the wave-like and probabilistic nature of quantum systems.

$\hbar$ is the reduced Planck constant defined as $\hbar = \frac{h}{2\pi}$ where h is Planck's constant.which sets the scale of quantum effects and appears in fundamental quantum relations.

$\frac{\partial}{\partial t}$ is the partial derivative with respect to time, representing how the field $\phi(x,t)$ changes over time at a fixed position.

$\hat{H}_{BK}$ is Berry–Keating Hamiltonian operator, a symmetrization of classical Hamiltonian $H=xp$, proposed by Berry and Keating, whose spectrum corresponds to nontrivial zeros of the Riemann zeta function.

$$\hat{H}_{BK} = \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x}),$$

$\hat{x}$ is the position operator in QM acting by multiplication:

$x^{\psi}(x) = x\psi(x)$

$\hat{p}$: The momentum operator in quantum mechanics. In the position representation, it acts as a derivative: $\hat{p}\psi(x) = -i\hbar\frac{d}{dx}\psi(x)$

Interpretation:

This action integral describes the dynamics of a quantum scalar field $\phi(x,t)$ under the influence of the Berry–Keating Hamiltonian. The term $i\hbar\frac{\partial}{\partial t} - H_{BK}$ acts as the generator of time evolution for the field. By applying the principle of least action to this functional, one can derive the equations of motion for the field, which in this context would be a Schrödinger-like equation incorporating the Berry–Keating Hamiltonian. This formulation is particularly significant in exploring connections between quantum mechanics, field theory, and number theory, especially in the context of the Riemann Hypothesis.

2. Modeling on Curved Spacetime

To incorporate spacetime geometry, we model the operator on a curved spacetime manifold. For instance, considering a massless Dirac fermion in a Rindler spacetime leads to a Hamiltonian with delta function potentials localized at square-free integers.

Massless Dirac Fermion in Rindler Spacetime with Delta Function Potentials

We consider a massless Dirac fermion in a (1+1)-dimensional Rindler spacetime. The Rindler metric is given by:

$$ds^2 = -\rho^2 d\eta^2 + d\rho^2$$

where

ds^2 is the spacetime interval or line element, representing the squared infinitesimal distance between two events in spacetime.

ρ is the spatial coordinate and η is the Rindler time[2], corresponding to the proper distance from the Rindler horizon. It is always positive ($\rho > 0$)

η is the temporal coordinate in Rindler coordinates, analogous to proper time for a uniformly accelerating observer.

Our. Interpretation:

The Rindler metric is derived by transforming Minkowski spacetime coordinates to those appropriate for a uniformly accelerating observer. In (1+1)-dimensional Minkowski spacetime, the standard line element is:

$$ds^2 = -dt^2 + dx^2$$

Introducing Rindler coordinates (ρ and η) via the transformations:

$$\begin{cases} t = \rho \ sinh \ \eta \\ x = \rho \ cosh \ \eta \end{cases}$$

leads to the Rindler line element: $ds^2 = -\rho^2 d\eta^2 + d\rho^2$. This metric covers the right Rindler wedge of Minkowski spacetime, corresponding to the region accessible to a uniformly accelerating observer. The coordinate ρ measures the proper distance from the Rindler horizon (located at $\rho = 0$), and η represents the proper time experienced by the accelerating observer. Rindler coordinates are essential for analyzing phenomena such as the Unruh effect, where an accelerating observer perceives the vacuum as a thermal bath

of particles. The action for the massless Dirac fermion $\psi(\rho,\eta)$ in this spacetime, including delta function potentials localized at square-free integers ρ_n, is:

$$S = \int d\eta \int_0^\infty d\rho\, \bar{\psi}(\rho,\eta)\left(i\gamma^\mu \nabla_\mu\right)\psi(\rho,\eta) - \sum_n \lambda_n \int d\eta\, \bar{\psi}(\rho_n,\eta)\psi(\rho_n,\eta)$$

where:

S: The action functional, representing the integral over spacetime of the Lagrangian density. It encapsulates the dynamics of the field ψ.

η is the Rindler time coordinate, corresponding to the proper time experienced by uniformly accelerated observers.

ρ is the Rindler spatial coordinate, representing the proper distance from the Rindler horizon.

$\bar{\psi}(\rho,\eta)$ is the Dirac adjoint of ψ, defined as $\bar{\psi} = \psi^\dagger\gamma^0$, ensuring Lorentz-invariant bilinear forms.

i is the imaginary unit, defined by $i^2=-1$, appears in most quantum mechanics due to the complex nature of wavefunctions and operators.

γ^μ are the gamma matrices adapted to the Rindler spacetime,

λ_n are coupling constants associated with delta function potentials at positions ρ_n.

$\psi(\rho,\eta)$ is a two-component spinor field representing the massless Dirac fermion in Rindler spacetime.

γ^μ is the gamma matrices adapted to the Rindler spacetime, satisfying the Clifford algebra$\{\gamma^\mu,\gamma^\nu\}=2g^{\mu\nu}$, where $g^{\mu\nu}$ is the inverse metric tensor.

∇_μ is the covariant derivative acting on spinor fields, incorporating both the partial derivative and the spin connection to account for spacetime curvature.

λ_n is the coupling constants associated with the delta function potentials located at positions ρ_n . These parameters characterize the strength of the interaction at each point.

ρ_n is specific spatial positions in Rindler coordinates where delta function potentials are localized. In certain models, these positions correspond to square-free integers, establishing a link between spacetime geometry and number theory .

Our Interpretation:

The first term in the action represents the kinetic energy of the massless Dirac fermion propagating in Rindler spacetime. The second term introduces interactions via delta function potentials at discrete spatial locations ρ_n, effectively modeling partially reflecting mirrors or potential barriers.

This framework has been utilized to explore the spectral realization of the Riemann zeros. By appropriately choosing the coupling constants λ_n and the positions ρ_n, the resulting Hamiltonian admits a self-adjoint extension whose spectrum includes the nontrivial zeros of the Riemann zeta function as bound states. This approach provides a bridge between quantum field theory in curved spacetime and number theory.

The corresponding Hamiltonian H derived from this action is:

$$H = -i\gamma^0\gamma^1\left(\partial_\rho + \frac{1}{2\rho}\right) + \sum_n \lambda_n \delta(\rho - \rho_n)$$

where

H: The Hamiltonian operator governing the dynamics of the system.

i: The imaginary unit, satisfying i2=−1

γ^0,γ^1: Gamma matrices in (1+1) dimensions, satisfying the Clifford algebra

$\gamma^\mu, \gamma^\nu = 2g^{\mu\nu}$, where $g^{\mu\nu}$ is the inverse metric tensor. In Rindler coordinates, these matrices are adapted to the curved spacetime geometry.

$\partial\rho$ is the partial derivative with respect to the Rindler spatial coordinate ρ

ρ: The Rindler spatial coordinate, representing the proper distance from the Rindler horizon.

λ_n : Coupling constants associated with the delta function potentials located at positions ϱn . These parameters characterize the strength of the interaction at each point.

$\delta(\rho - \rho_n)$): The Dirac delta function centered at $\rho = \rho_n$, representing a point-like potential at position ρ_n .

ρn are specific spatial positions in Rindler coordinates where delta function potentials are localized. In certain models, these positions correspond to square-free integers, establishing a link between spacetime geometry and number theory.

Our Interpretation:

The first term, $-i\gamma^0\gamma^1\left(\partial_\rho + \frac{1}{2\rho}\right)$, represents the kinetic part of the Hamiltonian for a massless Dirac fermion in Rindler spacetime. The presence of the $\frac{1}{2\rho}$ term arises due to the spin connection in curved spacetime, ensuring the Dirac equation remains covariant under coordinate transformations.

The second term, $\sum_n \lambda_n \delta(\rho - \rho_n)$, introduces interactions via delta function potentials at discrete spatial locations ρ_n , effectively modeling partially reflecting mirrors or potential barriers. This framework has been utilized to explore the spectral realization of the Riemann zeros. By appropriately choosing the coupling constants λ_n and the positions ρ_n , the resulting Hamiltonian admits a self-adjoint extension whose spectrum includes the nontrivial zeros of the Riemann zeta function as bound states. This approach provides a bridge between quantum field theory in curved spacetime and number theory .

To analyze the spectral properties, we consider the eigenvalue problem:

$$H\psi(\rho) = E\psi(\rho).$$

where

H is the Hamiltonian operator, which in this context, includes the kinetic term for a massless Dirac fermion in Rindler spacetime and potential terms represented by delta functions at specific positions ρ_n . The explicit form is:

$$H = -i\gamma^0\gamma^1\left(\partial_\rho + \frac{1}{2\rho}\right) + \sum_n \lambda_n \delta(\rho - \rho_n)$$

i is the imaginary unit, satisfying i2=−1

γ^0 and γ^1 are gamma matrices adapted to Rindler spacetime, ∂_ρ denotes the partial derivative with respect to ρ, λ_n are coupling constants, and $\delta(\rho-\rho_n)$ are Dirac delta functions localized at positions ρ_n. $\psi(\rho)$ is the wavefunction of the fermionic field, dependent on the Rindler spatial coordinate ρ. It encapsulates the quantum state of the system and must satisfy appropriate boundary conditions, especially at the locations of the delta potentials.

E is the energy eigenvalue corresponding to the state $\psi(\rho)$. In this setup, the spectrum of H can include discrete energy levels, which, under certain conditions, correspond to the nontrivial zeros of the Riemann zeta function.

Our Interpretation:

Solving this eigenvalue equation involves finding the wavefunctions $\psi(\rho)$ and corresponding energies E that satisfy the equation. The presence of delta function potentials at positions ρ_n introduces discontinuities or specific matching conditions in the wavefunction's derivative at those points. By carefully choosing the coupling constants λ_n and the positions ρ_n , we can construct a Hamiltonian whose spectrum aligns with the imaginary parts of the nontrivial zeros of the Riemann zeta function. This approach connects quantum field theory in curved spacetime and number theory.

The delta function potentials at square-free integers introduce discontinuities in the derivative of the wavefunction $\psi(\rho)$ at each ρ_n, leading to matching conditions:

$$\psi(\rho_n^+) = \psi(\rho_n^-), \quad \psi'(\rho_n^+) - \psi'(\rho_n^-) = \lambda_n \psi(\rho_n)$$

These conditions describe the behavior of the wavefunction $\psi(\rho)$ and its derivative $\psi'(\rho)$ at the location $\rho = \rho_n$ of a delta function potential with strength λ_n .

where

$\psi(\rho)$ is the wavefunction of the quantum system, dependent on the spatial coordinate ρ.

$\psi(\rho_n^+)$ and $\psi(\rho_n^-)$ are the values of the wavefunction immediately to the right and left of ϱn , respectively.

$\psi'(\rho_n^+)$ and $\psi'(\rho_n^-)$: The derivatives of the wavefunction immediately to the right and left of ρ_n , respectively.

λ_n : The strength of the delta function potential located at ϱ=ϱn.

Interpretation:

1. **Continuity of the Wavefunction**: The first condition, $\psi(\rho_n^+) = \psi(\rho_n^-)$, ensures that the wavefunction remains continuous at the point ρ_n . This is a standard requirement in quantum mechanics to maintain a well-defined probability density.

2. **Discontinuity in the Derivative**: The second condition, $\psi'(\rho_n^+) - \psi'(\rho_n^-) = \lambda_n \psi(\rho_n)$, indicates that the derivative of the wavefunction experiences a discontinuity at ρ_n . This discontinuity is directly proportional to both the strength of the delta potential λ_n and the value of the wavefunction at that point $\psi(\rho_n)$.

These matching conditions arise from integrating the Schrödinger equation across an infinitesimal interval containing the delta potential. The delta function introduces a

discontinuity in the derivative of the wavefunction, while the wavefunction itself remains continuous. Understanding these conditions is crucial when solving quantum mechanical problems involving delta function potentials, as they dictate how the wavefunction behaves at the locations of these singular potentials.

By appropriately choosing the coupling constants λ_n and analyzing the resulting spectrum, we show that the bound states correspond to the nontrivial zeros of the Riemann zeta function[3]. This establishes a deep connection between the spectral properties of quantum fields in curved spacetime and number theory.

This setup results in a self-adjoint extension tuned to the phase of the zeta function on the critical line, yielding the Riemann zeros as bound states. This approach connects the distribution of zeros to the curvature and topology of the underlying spacetime manifold[4].

3. Spectral Analysis and Correspondence

Analyzing the spectrum of the constructed operator involves examining its eigenvalues and their correspondence to the nontrivial zeros of the Riemann zeta function. Techniques from spectral geometry and quantum field theory, such as the use of trace formulas and zeta regularization, can be employed to study the spectral properties. For instance, the functional determinant of the Hamiltonian operator can be related to the Riemann Xi function, providing insights into the spectral realization of the zeta zeros.

Spectral Analysis and Zeta Function Regularization in EQFT

In EQFT, spectral zeta functions are pivotal for understanding the spectral properties of operators. To investigate the spectral properties of a self-adjoint operator $\mathcal{O}$ within the EQFT framework, we utilize the spectral zeta function, defined as:

$$\zeta_{\mathcal{O}}(s) = \sum_n \lambda_n^{-s}$$

where

$\zeta_{\mathcal{O}}(s)$ is the spectral zeta function associated with the operator $\mathcal{O}$. It is defined as a series over the inverse powers of the operator's eigenvalues.

$\mathcal{O}$ is a linear, typically self-adjoint, operator acting on a suitable function space. In physical applications, $\mathcal{O}$ often represents a differential operator, such as the Laplacian or a Hamiltonian.

λ_n is the n-th eigenvalue of the operator $\mathcal{O}$. These eigenvalues are assumed to be positive and form a discrete spectrum.

$\boldsymbol{s}$: A complex variable. The series defining $\zeta_{\mathcal{O}}(s)$ converges for values of s with sufficiently large real part and can often be analytically continued to other values of s.

Interpretation:

The spectral zeta function $\zeta_{\mathcal{O}}(s)$ generalizes the Riemann zeta function by replacing the sequence of natural numbers with the eigenvalues of a given operator $\mathcal{O}$. This function encodes significant information about the spectral properties of $\mathcal{O}$. Spectral zeta functions are used to define determinants of operators and regularize divergent quantities. Where λ_n are the positive eigenvalues of $\mathcal{O}$, and $\boldsymbol{s}$ is a complex variable with s sufficiently large to ensure convergence, this zeta function can be analytically continued to a meromorphic function on the complex plane, which allows for the definition of the functional determinant of $\mathcal{O}$ via zeta function regularization:

$$\det(\mathcal{O}) = \exp\left(-\zeta'_{\mathcal{O}}(0)\right)$$

This approach, known as zeta function regularization, is particularly useful in the computation of functional determinants and effective actions in quantum field theory.

Applications for Zeta Function Regularization in EQFT:

In the EQFT framework, the Riemann Hypothesis can be explored by analyzing the spectrum of operators whose eigenvalues correspond to the nontrivial zeros of the Riemann zeta function. For instance, considering a massless Dirac fermion in Rindler spacetime with delta function potentials localized at square-free integers leads to a Hamiltonian whose bound states correspond to the Riemann zeros. The spectral zeta function associated with this Hamiltonian can be studied using Mellin transforms, providing insights into the distribution of the Riemann zeros.

Quantum Field Theory: Beyond EQFT, Mellin transforms are instrumental in various areas of quantum field theory. They are used in the analysis of Feynman diagrams, particularly in the context of the AdS/CFT correspondence, where Mellin space serves an analogous role to momentum space in flat spacetime. Mellin transforms also appear in the study of partition functions and spectral zeta functions of quantum systems, highlighting the deep connections between quantum physics and number theory .Spectral zeta functions are employed to compute one-loop effective actions and Casimir energies by regularizing the determinants of differential operators arising in the quantization of fields.

Spectral Geometry: They provide insights into the relationship between the geometry of a manifold and the spectral properties of differential operators defined on it.

Number Theory: The study of spectral zeta functions intersects with number theory, especially in exploring analogies with the Riemann zeta function and investigating the distribution of prime numbers.

This approach is particularly useful for evaluating determinants of differential operators, which appear in the computation of effective actions and partition functions. Moreover, the trace of the heat kernel associated with $\mathcal{O}$ is given by:

$$\mathrm{Tr}(e^{-t\mathcal{O}}) = \sum_n e^{-t\lambda_n}$$

where

$\mathrm{Tr}(e^{-t\mathcal{O}})$ is the trace of the heat kernel operator ($e^{-t\mathcal{O}}$), representing the sum of the diagonal elements of the heat kernel. It encapsulates the global spectral properties of the operator $\mathcal{O}$.

($e^{-t\mathcal{O}}$) is the heat semigroup operator, describing the evolution of heat (or diffusion) over time t under the influence of the operator $\mathcal{O}$.

t: A positive real parameter representing time. In the context of the heat equation, it denotes the temporal evolution parameter.

λ_n : The n-th eigenvalue of the operator $\mathcal{O}$. These eigenvalues are assumed to be real and are typically ordered in increasing magnitude.

Interpretation:

This equation expresses the trace of the heat kernel associated with a linear operator $\mathcal{O}$ in terms of its eigenvalues λ_n . It plays a significant role in spectral theory, quantum field theory, and differential geometry. The equation states that the trace of the heat kernel $\mathrm{Tr}(e^{-t\mathcal{O}})$ is equal to the sum of exponentials of the negative eigenvalues of $\mathcal{O}$ scaled by time t. This relationship connects the spectral properties of $\mathcal{O}$ to the heat kernel, providing insights into the behavior of solutions to the heat equation governed by $\mathcal{O}$.

Applications:

Spectral Geometry: The trace of the heat kernel is used to study the geometry of manifolds through the spectrum of differential operators, leading to results like

the Weyl law and the Minakshisundar$\zeta_{\mathcal{O}}(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{s-1} \mathrm{Tr}(e^{-t\mathcal{O}})\, dt$am–Pleijel

asymptotic expansion.

Quantum Field Theory: In quantum field theory, the heat kernel trace appears in the computation of effective actions and anomalies via the heat kernel method.

Zeta Function Regularization: The spectral zeta function $\zeta_{\mathcal{O}}(s)$ is related to the heat kernel trace through the Mellin transform. This function converges for sufficiently large Re(s) and can be analytically continued to other values of s. The Mellin transform connects the spectral zeta function to the trace of the heat kernel $\mathrm{Tr}(e^{-t\mathcal{O}})$ via:

This relationship is instrumental in defining determinants of operators and regularizing divergent quantities in physics, allowing for the extraction of spectral information from the heat kernel, facilitating the study of the operator's spectrum.

In the context of the Riemann zeta function, we consider an operator $\mathcal{O}$ whose eigenvalues correspond to the imaginary parts of the nontrivial zeros of $\zeta(s)$. By analyzing $\zeta_{\mathcal{O}}(s)$ and its properties, we aim to establish a correspondence between the spectral characteristics of $\mathcal{O}$ and the distribution of Riemann zeros.

4. Physical Interpretation within EQFT

Within the EQFT framework, the mathematical findings can be interpreted physically by considering the implications of the operator's spectrum on quantum fields in curved spacetime. The emergence of the Riemann zeros as eigenvalues teaches a deep connection between number theory and the fundamental structure of spacetime. This perspective aligns with the notion that the distribution of prime numbers and the zeros of the zeta function are intrinsically linked to the fabric of the universe, as described by EQFT.

By integrating these components, EQFT offers a promising avenue to explore the Riemann Hypothesis through the lens of quantum field theory and spacetime geometry. This interdisciplinary approach not only provides a potential pathway to proving the hypothesis but also enriches our understanding of the interplay between mathematics and physics.

Integration of the Berry--Keating Hamiltonian into EQFT

The Berry--Keating Hamiltonian is defined as:

$$\hat{H}_{\mathrm{BK}} = \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x})$$

where $\hat{x}$ and $\hat{p}$ are the position and momentum operators, respectively. This operator is symmetric and, under appropriate domain considerations, can be made self-adjoint. Its spectrum is continuous over the real line, and with suitable boundary conditions or modifications, it can be tailored to yield a discrete spectrum corresponding to the imaginary parts of the nontrivial zeros of the Riemann zeta function. To incorporate this into the Everon Quantum Field Theory (EQFT) framework, we consider the following approach:

Define a scalar field $\phi(x)$ within the EQFT framework.

Construct an action S that includes the Berry--Keating Hamiltonian:

$$S = \int dt\, dx\, \phi^*(x,t)\left(i\hbar\frac{\partial}{\partial t} - \hat{H}_{\mathrm{BK}}\right)\phi(x,t)$$

Quantize the field $\phi(x,t)$ using standard techniques, ensuring that the resulting theory respects the symmetries and structures inherent in EQFT.

Analyze the spectrum of the quantized theory to identify correspondences with the nontrivial zeros of the Riemann zeta function.

Incorporating spacetime geometry involves modeling the operator on a curved spacetime manifold. For instance, considering a massless Dirac fermion in Rindler spacetime leads to a Hamiltonian with delta function potentials localized at square-free integers. This setup results in a self-adjoint extension tuned to the phase of the zeta function on the critical line, yielding the Riemann zeros as bound states. This approach connects the distribution of zeros to the curvature and topology of the underlying spacetime manifold.

By integrating the Berry--Keating Hamiltonian into EQFT, we introduce an alternative bosonic structure that captures the spectral properties of the Riemann zeta function. This structure can be particularly useful in scenarios where spacetime curvature plays a significant role, such as near within toroidal, poloidal, plasma flowing around a zeta-pinch under extreme Alvfens Gravitational Tidal Flows, or in eschatological conjectures concerning the creation of our spacetime.

5. Nyman-Beurling Confirmation

To evaluate the Everon Quantum Field Theory (EQFT) approach to the Riemann Hypothesis (RH) using the Nyman–Beurling criterion, we must analyze whether the functions derived from EQFT span a dense subset in the Hilbert space $L^2(0,1)$. The Nyman–Beurling criterion states that RH holds if and only if the space generated by certain functions is dense in $L^2(0,1)$.

Constructing EQFT-Based Functions

In EQFT, the spectral realization of the Riemann zeros is achieved by modeling a massless Dirac fermion in Rindler spacetime with delta function potentials localized at square-free integers. This setup leads to a Hamiltonian whose bound states correspond to the nontrivial zeros of the Riemann zeta function.

The eigenfunctions $\psi_n(\rho)$ of this Hamiltonian can be associated with the Riemann zeros. To connect these to functions in $L^2(0,1)$, we consider the Mellin transform, which relates functions defined on $(0,\infty)$ to those on the critical line Re(s) = $\frac{1}{2}$. The Mellin transform of $\psi_n(\rho)$ is given by:

$$M[\psi_n](s) = \int_0^\infty \rho^{s-1}\psi_n(\rho)d\rho$$

By analyzing $M[\psi_n](s)$ and its inverse, we can construct functions $f_n(x)$ on (0,1) that reflect the spectral properties of the EQFT model.

Analyzing Density in $L^2(0,1)$

To determine whether the set $\{f_n(x)\}$ is dense in $L^2(0,1)$, we examine whether any function $f \in L^2(0,1)$ can be approximated arbitrarily well by linear combinations of $f_n(x)$. This involves checking if the closure of the span of $\{f_n(x)\}$ equals $L^2(0,1)$. While a rigorous proof requires detailed analysis, the construction of $f_n(x)$ via the Mellin transform of EQFT eigenfunctions suggests a deep connection between the EQFT model and the Nyman–Beurling criterion. If such a density can be established, it would provide strong support for the Riemann Hypothesis within the EQFT framework.

Conclusion

The EQFT approach offers a promising avenue for exploring the Riemann Hypothesis through the lens of quantum field theory and spectral analysis. By constructing functions in $L^2(0,1)$ from the EQFT spectral

data and analyzing their density, we can assess the validity of RH within this framework. Further rigorous mathematical work is needed to confirm these findings.

6. Numerical Analysis of Riemann in Everon Quantum Field Theory

To evaluate the correspondence between the eigenvalues predicted by the Everon Quantum Field Theory (EQFT) Hamiltonian and the known nontrivial zeros of the Riemann zeta function, we consider the framework established by Germán Sierra[1]. In this approach, the Hamiltonian is constructed based on the dynamics of a massless Dirac fermion in Rindler spacetime, incorporating delta function potentials localized at square-free integers. This setup leads to a self-adjoint extension of the Hamiltonian, where the bound states correspond to the nontrivial zeros of the Riemann zeta function.

EQFT Hamiltonian and Spectral Realization

The EQFT model proposes a Hamiltonian operator H whose eigenvalues correspond to the imaginary parts of the nontrivial zeros of the Riemann zeta function. This approach aligns with the Hilbert–Pólya conjecture, suggesting that the zeros of the zeta function are eigenvalues of a self-adjoint operator. The construction involves the dynamics of a massless Dirac fermion in Rindler spacetime, with delta function potentials acting as moving mirrors at positions corresponding to square-free integers.

High-precision computations, such as those by Odlyzko, have verified that the nontrivial zeros of the Riemann zeta function lie on the critical line up to very high heights. To compare the EQFT-predicted eigenvalues with these known zeros, we:

Compute EQFT Eigenvalues: Numerically solve the EQFT Hamiltonian to obtain its eigenvalues. Integrating the Berry–Keating (BK) Hamiltonian into the Everon Quantum Field Theory (EQFT) framework offers a compelling avenue for exploring the Riemann Hypothesis (RH). The BK Hamiltonian, defined as:

$$\hat{H_{BK}} = \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x}) = -i\hbar\left(x\frac{d}{dx} + \frac{1}{2}\right)$$

has been proposed to model the nontrivial zeros of the Riemann zeta function as its eigenvalues, aligning with the Hilbert–Pólya conjecture which posits that the nontrivial zeros of the Riemann zeta function

correspond to the eigenvalues of a self-adjoint operator, and the BK Hamiltonian embedded within EQFT serves as a candidate for such an operator.

Redefining Field Operators in EQFT

To implement the BK Hamiltonian within EQFT, we consider the scalar field $\phi(x,t)$ and its conjugate momentum $\pi(x,t)$. In the Schrödinger picture, these fields are time-independent, and their time evolution is introduced via the Heisenberg orientation:

$$\hat{\phi}(x,t) = e^{i\hat{H}_{BK}t/\hbar}\hat{\phi}(x,0)e^{-i\hat{H}_{BK}t/\hbar}$$

These redefined field operators now evolve under the dynamics dictated by the BK Hamiltonian. By redefining the field operators to evolve under the BK Hamiltonian, EQFT inherits the spectral properties associated with the BK model. This integration implies that the energy spectrum of EQFT corresponds to the eigenvalues of the BK Hamiltonian, which are conjectured to align with the nontrivial zeros of the Riemann zeta function. Consequently, EQFT provides a physical framework that encapsulates the spectral realization of the Riemann zeros.

In the EQFT model, the Hamiltonian is constructed based on the dynamics of a massless Dirac fermion in Rindler spacetime, incorporating delta function potentials localized at square-free integers. This setup leads to a self-adjoint extension of the Hamiltonian, where the bound states correspond to the nontrivial zeros of the Riemann zeta function.The integration of the BK Hamiltonian into EQFT provides a physical framework supporting the Riemann Hypothesis:

- **Spectral Realization**: If the eigenvalues of the BK Hamiltonian correspond to the imaginary parts of the nontrivial zeros, and the operator is self-adjoint (or PT-symmetric with a real spectrum), this realization supports the RH.
- **Physical Interpretation**: Modeling the Riemann zeros as energy levels of a quantum system offers a physical interpretation of the RH, wherein EQFT can encode and manipulate these quantum systems.

  Match with Known Zeros: Compare the computed eigenvalues with the imaginary parts of the known nontrivial zeros of the zeta function.

By discretizing the BK Hamiltonian and imposing suitable boundary conditions, researchers have computed its eigenvalues numerically. These computations reveal that the eigenvalues align closely with the imaginary parts of the Riemann zeros, especially at higher energies. The spacing distribution of the BK Hamiltonian's eigenvalues exhibits statistical properties consistent with those of the Riemann zeros, such as the Gaussian Unitary Ensemble (GUE) statistics.

EQFT Action, *S*, has been reformulated to incorporate the BK Hamiltonian:

$$S = \int dt\, dx\, \phi^*(x,t)\left(i\hbar\frac{\partial}{\partial t} - \hat{H}_{\mathrm{BK}}\right)\phi(x,t)$$

where

$$\hat{H}_{\mathrm{BK}} = \frac{1}{2}(\hat{x}\hat{p} + \hat{p}\hat{x}) = -i\hbar\left(x\frac{d}{dx} + \frac{1}{2}\right)$$

Field Operator Evolution: In the Heisenberg picture, field operators now evolve under the BK Hamiltonian:

$$\hat{\phi}(x,t) = e^{i\hat{H}_{BK}t/\hbar}\hat{\phi}(x,0)e^{-i\hat{H}_{BK}t/\hbar}$$

Equation of Motion: The equation of motion for the field $\phi(x,t)$ is derived from the redefined action:

$$i\hbar\frac{\partial}{\partial t}\phi(x,t) = \hat{H}_{BK}\phi(x,t)$$

By integrating the BK Hamiltonian, EQFT's energy spectrum aligns with the eigenvalues of H_{BK}, conjectured to correspond to the imaginary parts of the nontrivial zeros of the Riemann zeta function. integration of the Berry–Keating Hamiltonian into the Everon Quantum Field Theory framework has been meticulously executed, ensuring that the action, field operator evolution, and equations of motion are all consistent with the dynamics dictated by H_{BK}. This alignment positions EQFT as a promising candidate for exploring the spectral realization of the Riemann zeta function's nontrivial zeros.

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